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INFUSED SOLID FUEL FOR HYBRID ROCKETS AND ORDNANCE

Abstract

An infused solid polymeric fuel for use in a hybrid rocket or as explosive ordnance. An infused solid polymeric fuel can include a porous polymeric body having continuous pores distributed throughout a volume of the polymeric body and connecting to an external surface of the polymeric body. A combustion enhancement agent can be infused into the polymeric body. The combustion enhancement agent can be deposited on interior surfaces of the pores and on the external surface of the polymeric body. The combustion enhancement agent can include a catalyst, an oxidizer, a hypergolic fuel, or a combination.

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Background/Summary

RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Patent Application No. 63/561,675, filed Mar. 5, 2024 which is incorporated herein by reference.

BACKGROUND

[0003] Significant advances in the miniaturization of electronic components have emerged during the first two decades of the 21st century and have allowed spacecraft bus sizes to shrink by nearly an order of magnitude. Spacecraft as small as 25 kg now offer the sensing and computational capability of a spacecraft weighing more than several hundred kilograms from just a generation ago. As a result, global interests in very small spacecraft (SmallSats) have grown dramatically, and a competitive commercial market has emerged during the last decade.

[0004] To date, SmallSats have primarily been used for educational, technology demonstration, or other novelty purposes. SmallSats are currently launched using a limited number of “ride-share” options that rely on the excess launch vehicle payload capacity. These ride-share options offer secondary payload satellite customers little or no control over the launch schedule and the achieved final orbit. To support the expanding capability of SmallSats, a cheap, reliable, high-performing, safe, and preferably, “green” propulsion unit can be useful in order to re-position the spacecraft properly; and once-inserted, maintain the required orbit. Unfortunately, SmallSat technology development has primarily centered on spacecraft bus design and miniaturization of sensor components. Generally, the propulsion industry has not kept pace with the bus avionics growth trend. With the current state of the art only a few space propulsion options are available to the systems designer. The cost associated with these propulsion systems has led to many small satellites launching with no propulsion: often restricting the capability of the mission.

[0005] Hydrazine is a very effective and reliable monopropellant. Hydrazine thrusters are simple, versatile, and dependable. Hydrazine technology has been under development since the late 1950's and has a very high technology readiness level (TRL) level. Unfortunately, hydrazine-based systems bring a wide range of objective and environmental hazard issues. Hydrazine is a highly toxic and dangerously unstable substance. The United States Environmental Protection Agency classifies hydrazine as “highly toxic.” Hydrazine is destructive to living tissues and is a known carcinogen. Hydrazine possesses a high vapor pressure and boils readily at room temperature. Thus, there exists significant risk of respiratory contamination, and special servicing procedures that employ full-pressure self-contained atmospheric protective ensemble suits are used. The toxicity and explosion potential of hydrazine leads to the use of extreme handling precautions.

[0006] Although some low-TRL electrical-chemical systems in SmallSat scales have been tested, currently the only operational green chemical-propellant options are associated with the ionic-liquids (ILs) Ammonium Dinitramide (ADN) and Hydroxyl Ammonium Nitrate (HAN). Both propellants have achieved at least one spaceflight as of early 2024, with the ADN-based propellant LMP-103S flying on the PRISMA Spacecraft, and the HAN-based propellant AFM-315E flying on NASA's Green Propellant Infusion Mission (GPIM). The PRISMA flight report states that LMP-103S delivered equivalent-to-superior performance to hydrazine, with a vacuum specific impulse of 230 seconds. The GPIM flight report does not specifically report the achieved in-flight specific impulse (Isp) for AFM-315E, but extrapolations from ground tests show that Isp values between 220 and 230 seconds are achievable.

[0007] In spite of being called “green,” by their manufacturers, IL-based propellants are generally

not environmentally-friendly. Both of the above-mentioned IL-propellants are toxic to organic tissue, for example AF-M315E contains hydroxyethyl hydrazine (HEHN) as an ignition enhancer and burn stabilizer. In high concentrations IL-based propellants are prone to energetic, uncontrolled decomposition events. Thus, special handling precautions are required for processing and storage. The major advantage of both propellants is a low-vapor pressure at room temperature, allowing handling and servicing without the use of Self Contained Atmospheric Protective Ensemble (SCAPE) suits.

[0008] Because of these properties, the USAF has recently begun to refer to such IL-formulations more properly as having “reduced toxicity” instead of being “green.” In addition to potential toxicity and objective hazards, there exist several key developmental issues that make IL-based propellants unsuitable for some small spacecraft applications.

[0009] The high water content makes IL-propellants notoriously hard to ignite, and LMP-103S and AFM-315E propellants cannot be cold-started. Multiple catalyst systems have been developed to augment IL ignitability; however, the associated catalyst beds must be preheated from 350-400° C. before and during ignition. This pre-heat cycle consumes a significant amount of energy. For example, on the Prisma demonstration flight the pre-heat time was 30 min, requiring 16.7 kJ of total energy input. This input was required for every cold-cycle of the LMP-103S thruster. Thus, it can be concluded that the ionic liquid catalyst beds and associated heating systems add significantly to the inert mass of the spacecraft and the high-wattage preheat requirement presents a significant disadvantage for small spacecraft where power budgets are extremely limited.

[0010] Also, due to the very slow reaction kinetics demonstrated by Hori and Katsumi for HAN, and Whitmore and Burnside for ADN at the moderate pressures levels required for SmallSat Thruster systems (1000-2000 kPa), ignition latencies can be significant, up to several seconds, and may limit the usefulness of IL-propellants for spacecraft maneuvering and control systems. Below 1000 kPa chamber pressure the IL liquid propellants will not ignite, and this issue limits the ability to scale these systems for thrusters below the 1-N thrust level.

[0011] Clearly, technology improvements must occur before IL-based systems can be employed as a primary propulsion unit or as part of the reaction control system for small spacecraft. With the current state of propulsion technology, the only proven non-hazardous propulsion alternative to hydrazine, and available for small ride-share payloads, is based on low-performing cold-gas thrusters.

[0012] Even under ideal operating conditions, when compared to conventional solid, hybrid, or bi-propellant options, the performances of ionic liquid propellants are generally quite low with achieved vacuum Isp values at or less than 230 seconds. The combination of these middling performance characteristics and lack of truly “green” advantages have led some in the industry to question whether ionic liquids as propellants have been “oversold”. Clearly, significant technological issues exist and must be overcome before the ionic-liquid propellants are used routinely as a hydrazine replacement.

[0013] In response to this emerging need for environmentally-sustainable in-space propulsion systems, hybrid rocket systems have emerged as a potential alternative to traditional toxic-propellants. When compared to conventional liquid- and solid-propelled rocket systems, hybrid rockets possess well-known operational safety and handling-advantages. A study by the U.S. Department of Transportation concluded that hybrid rocket motors can be safely stored and operated without a significant risk of explosion or detonation and offer the potential to significantly reduce operating costs for commercial launch vehicles. Finally, hybrid rocket systems offer higher performance than hydrazine-based systems; and their inherent-design safety offers a significant potential for ride-share spacecraft applications.

[0014] However, in spite of these well-known safety and handling advantages; conventionally-designed hybrid rocket systems have not seen widespread commercial use due to two key technical challenges. The first technical challenge is the current lack of an operational method reliable

multiple-use ignition, allowing on-demand start, stop, and re-start. The second technical challenge is low fuel regression rates associated with hybrid combustion.

[0015] Because of the relative propellant stability, hybrid rocket systems can be difficult to ignite; and a substantial ignition enthalpy source is required. The ignition source must provide sufficient heat to pyrolyze the solid fuel grain at the head end of the motor, while simultaneously providing sufficient residual energy to overcome the activation energy of the propellants. Such high-energy devices often come with a suite of environmental and objectives risks, and operational challenges.

[0016] Most conventional hybrid rocket applications have used high output pyrotechnic or “squib” charges to initiate combustion. Pyrotechnic charges are extremely susceptible to inadvertent detonation due to electromagnetic radiation, and present significant explosion hazards that are incompatible with rideshare opportunities. Most importantly, for nearly all applications pyrotechnic ignitors are designed as “one-shot” devices that do not allow a multiple restart capability. Thus, the great potential for restartable upper stages or in-space maneuvering systems using hybrid propulsion has remained largely unrealized. An operational hybrid system with multiple restart capability does not currently exist.

[0017] The second disadvantage is that the internal motor ballistics of hybrid combustion produce regression rates that are typically between 30 and 40% of the regression rates of solid fuel motors of the same thrust and impulse class. To make up for the lower regression rate, a higher oxidizer mass flow rate is required to maintain the same thrust level. This increases the system's oxidizer-to-fuel (O/F) ratio and ultimately results in poor mass impulse performance, erosive fuel burning, nozzle erosion, reduced motor duty cycles, potential combustion instability, and poorer overall performance of the system. To overcome this O/F problem, hybrid rockets are traditionally designed with cylindrical fuel ports that have high length-to-diameter ratios. This high aspect ratio can result in poor volumetric efficiency, and thus limits a hybrid motor application to a customarily volume-constrained small satellite.

SUMMARY

[0018] This disclosure describes infused solid polymeric fuels. In one example, an infused solid polymeric fuel can include a porous polymeric body having continuous pores distributed throughout a volume of the polymeric body and connecting to an external surface of the polymeric body. A combustion enhancement agent can be infused into the polymeric body. The combustion enhancement agent can be deposited on interior surfaces of the pores and on the external surface of the polymeric body. The combustion enhancement agent can include a catalyst, an oxidizer, a hypergolic fuel, or a combination thereof.

[0019] The present disclosure also describes methods of making infused solid polymeric fuels. In one example, a method of making an infused solid polymeric fuel can include forming a porous polymeric body having continuous pores distributed throughout a volume of the polymeric body and connecting to an external surface of the polymeric body. The porous polymeric body can be soaked in a composition comprising a combustion enhancement agent and a solvent to deposit the combustion enhancement agent on interior surfaces of the pores and on the external surface of the polymeric body. The solvent can be evaporated. The combustion enhancement agent can include a catalyst, an oxidizer, a hypergolic fuel, or a combination thereof.

[0020] The present disclosure also describes hybrid rocket systems. In one example, a hybrid rocket system can include an infused solid polymeric fuel grain. The infused solid polymeric fuel grain can include a porous polymeric body having continuous pores distributed throughout a volume of the polymeric body and connecting to an external surface of the polymeric body, and a decomposition catalyst infused into the polymeric body, wherein the catalyst is deposited on interior surfaces of the pores and on the external surface of the polymeric body. The hybrid rocket system can also include a liquid oxidizer injector operable to contact the fuel grain with liquid oxidizer such that the catalyst catalyzes a decomposition of the liquid oxidizer. The system can also include an igniter operable to ignite the infused solid polymer fuel grain.

[0021] There has thus been outlined, rather broadly, the more important features of the invention so that the detailed description thereof that follows may be better understood, and so that the present contribution to the art may be better appreciated. Other features of the present invention will become clearer from the following detailed description of the invention, taken with the accompanying drawings and claims, or may be learned by the practice of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. 1 is a schematic illustration of a High-Performance Green Hybrid Propulsion ignition electronics system in accordance with an example of the present disclosure.

[0023] FIG. 2 is a schematic of the head-end of a 3D-printed ABS fuel grain with embedded electrodes in accordance with an example of the present disclosure.

[0024] FIG. 3 is a perspective view of an example infused polymeric solid fuel with an example spark cap in accordance with an example of the present disclosure.

[0025] FIG. 4A-4B are cross-sectional views of a portion of an example infused polymeric solid fuel in accordance with an example of the present disclosure.

[0026] FIG. 5 shows a motor configuration of an example hybrid rocket motor in accordance with an example of the present technology.

[0027] FIG. 6 is a schematic diagram of a test cart used to test an example hybrid rocket motor in accordance with an example of the present technology.

[0028] FIG. 7 is a side view of a thrust stand mounting system for testing the example hybrid rocket motor in accordance with an example of the present technology.

[0029] FIG. 8A-8D are graphs of testing results of an example hybrid rocket motor in accordance with an example of the present technology.

[0030] FIG. 9 shows a graph of comparison test results of a rocket motor without an infused catalyst.

[0031] FIG. 10 is a graph of thrust vs. time for a hybrid rocket motor in accordance with an example of the present technology.

[0032] These drawings are provided to illustrate various aspects of the invention and are not intended to be limiting of the scope in terms of dimensions, materials, configurations, arrangements or proportions unless otherwise limited by the claims.

DETAILED DESCRIPTION

[0033] While these exemplary embodiments are described in sufficient detail to enable those skilled in the art to practice the invention, it should be understood that other embodiments may be realized and that various changes to the invention may be made without departing from the spirit and scope of the present invention. Thus, the following more detailed description of the embodiments of the present invention is not intended to limit the scope of the invention, as claimed, but is presented for purposes of illustration only and not limitation to describe the features and characteristics of the present invention, to set forth the best mode of operation of the invention, and to sufficiently enable one skilled in the art to practice the invention. Accordingly, the scope of the present invention is to be defined solely by the appended claims.

Definitions

[0034] In describing and claiming the present invention, the following terminology will be used.

[0035] The singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a fuel grain” includes reference to one or more of such fuel grains and reference to “the injection port” refers to one or more of such injection ports.

[0036] As used herein with respect to an identified property or circumstance, “substantially” refers

to a degree of deviation that is sufficiently small so as to not measurably detract from the identified property or circumstance. The exact degree of deviation allowable may in some cases depend on the specific context.

[0037] As used herein, “adjacent” refers to the proximity of two structures or elements.

Particularly, elements that are identified as being “adjacent” may be either abutting or connected. Such elements may also be near or close to each other without necessarily contacting each other. The exact degree of proximity may in some cases depend on the specific context.

[0038] As used herein, the term “about” is used to provide flexibility and imprecision associated with a given term, metric or value. The degree of flexibility for a particular variable can be readily determined by one skilled in the art. However, unless otherwise enunciated, the term “about” generally connotes flexibility of less than 2%, and most often less than 1%, and in some cases less than 0.01%.

[0039] As used herein, a plurality of items, structural elements, compositional elements, and/or materials may be presented in a common list for convenience. However, these lists should be construed as though each member of the list is individually identified as a separate and unique member. Thus, no individual member of such list should be construed as a de facto equivalent of any other member of the same list solely based on their presentation in a common group without indications to the contrary.

[0040] As used herein, the term “at least one of” is intended to be synonymous with “one or more of.” For example, “at least one of A, B and C” explicitly includes only A, only B, only C, or combinations of each.

[0041] Numerical data may be presented herein in a range format. It is to be understood that such range format is used merely for convenience and brevity and should be interpreted flexibly to include not only the numerical values explicitly recited as the limits of the range, but also to include all the individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range is explicitly recited. For example, a numerical range of about 1 to about 4.5 should be interpreted to include not only the explicitly recited limits of 1 to about 4.5, but also to include individual numerals such as 2, 3, 4, and sub-ranges such as 1 to 3, 2 to 4, etc. The same principle applies to ranges reciting only one numerical value, such as “less than about 4.5,” which should be interpreted to include all of the above-recited values and ranges. Further, such an interpretation should apply regardless of the breadth of the range or the characteristic being described.

[0042] Any steps recited in any method or process claims may be executed in any order and are not limited to the order presented in the claims. Means-plus-function or step-plus-function limitations will only be employed where for a specific claim limitation all of the following conditions are present in that limitation: a) “means for” or “step for” is expressly recited; and b) a corresponding function is expressly recited. The structure, material or acts that support the means-plus function are expressly recited in the description herein. Accordingly, the scope of the invention should be determined solely by the appended claims and their legal equivalents, rather than by the descriptions and examples given herein.

Infused Solid Polymeric Fuel

[0043] In order to fill the technology gaps as described in the previous section, a High Performance Green Hybrid Propulsion (HPGHP) technology has been developed as a safe and environmentally-sustainable replacement for hydrazine across a wide range of applications.

[0044] HPGHP is enabled by recent advances in 3D printing and leverages unique electrical breakdown characteristics of certain 3D-printed plastics, the most effective being acrylonitrile butadiene styrene (ABS). Additive manufacturing changes the electrical breakdown properties, and when printed materials are presented with an inductive electrical potential, electrical-arcing along the layered surface pyrolyzes material and seeds combustion when an oxidizing flow is introduced. This “sparking” property has been developed into a proprietary, power-efficient system that can be

cold-started and restarted with a high degree of reliability. Multiple prototype units with thrust levels varying from less than 1 N to greater than 900 N have been developed and tested.

[0045] Although there may appear to be some similarities, the arc-ignition method described is distinctly different from the action of a pulse plasma thruster. Pulse plasma thruster designs use a high alternating current source to pyrolyze the surface of a Teflon block. The high current in the plasma arc induces a magnetic field. The action of the current and the magnetic field causes the plasma to be accelerated. When the current is stopped, pyrolysis ceases. A typical ignition cycle uses up to 1 kW of power. The arc-ignition method used in this work uses a low direct current source to pyrolyze the fuel material. Additive printing of the fuel changes the material dielectric proper ties; when ABS is subjected to the electro-static potential between embedded electrodes, the layered structure allows an “arc track” to be carved between the electrodes. Associated joule heating pyrolyzes the fuel; and as oxidizing flow is introduced, ignition spontaneously occurs. Combustion continues even after the current source is terminated. A typical ignition cycle uses less than 5 W of power applied for less than a second, and consumes less than 5 joules of total ignition energy. Once started, the system can be sequentially fired with no additional energy inputs required. FIG. 1 shows a schematic of the HPGHP Ignition System Electronics, and FIG. 2 shows the head-end of a 3D-printed ABS fuel grain with embedded electrodes. This system is described in U.S. Pat. No. 10,774,789 B2, which is hereby incorporated herein by reference.

[0046] The system has been scaled over a large range with successful prototypes with thrust levels varying from 0.5 to 900 N having been tested. Multiple oxidizers including gaseous oxygen (GOX), nitrous oxide (N_2O), hydrogen peroxide (H_2O_2), enriched air (EAN40), and Nitrox have been successfully tested with the HPGHP system. Nitrox is a “green” blend of GOX and N_2O and is similar to the laughing gas used for medical anesthesia applications. A flight-weight 25 N thruster system was extensively vacuum tested.

[0047] A flight experiment containing a 10 N prototype of this thruster system was launched aboard a two-stage Terrier-Improved Malemute sounding rocket from Wallops Flight Facility. The launch achieved apogee of 172 km, allowing more than 6 min in a hard-vacuum environment above the Von-Karman line. The thruster was successfully fired five times.

[0048] In its most mature form, the HPGHP system uses GOX as the oxidizer with 3D-printed ABS as the fuel. The GOX/ABS propellants are highly mass efficient system, with a flight weight 25 N thruster system achieving vacuum Isp greater than 300 seconds. Unfortunately, unless stored at very high pressures, GOX has a low specific gravity and is a volumetrically inefficient propellant.

[0049] Due to its high density, hydrogen peroxide was considered to be a very promising alternative oxidizer for this application. In order to achieve comparable density to H_2O_2 , GOX would need to be stored at pressures above 10,000 psi. When using a 90% hydrogen peroxide solution as an oxidizer along with a thermoplastic fuel in a hybrid system, there exists the potential for performance at a very high level.

[0050] Although HPGHP ignition works quite reliably using GOX as the oxidizer; HPGHP has experienced reliability and ignition latency issues when replaced by hydrogen peroxide. High concentrations of 90% or greater of hydrogen peroxide, referred to as high-test peroxide (HTP), have been used extensively for propulsion applications, both as a monopropellant and in combinations with fuels. Hydrogen peroxide decomposes to form water and oxygen gas. In this reaction both oxidation and reduction occur at the same time. This reaction is very energetic producing up to 98.1 kJ for every mole of peroxide that is decomposed (3.33 MJ/kg). Typically, an aqueous solution of HTP is sufficiently stable to work with, requiring an activation energy of approximately 75 KJ/mol in the absence of a catalyst.

[0051] In typical rocket applications with H_2O_2 as a monopropellant, a heated catalyst bed (catbed) is used to initiate decomposition. The catbed lowers the activation energy to the point where a moderate amount of heat can initiate decomposition. Noble metal catalysts like platinum or silver can lower the activation energy to less than 50 KJ/mol. Although catalytic decomposition of

monopropellant HTP has been successfully used for a variety of applications, this method typically requires very high concentrations of peroxide, greater than 90%. Even then, wet partially decomposed burns are very typical. Even in fully decomposed HTP plumes, water is an inherent by-product.

[0052] When catalytic decomposition is applied to hybrid rocket systems, the results are less satisfactory. As the decomposed oxidizer plume exits the injector and enters the hybrid combustion chamber, it rapidly expands and super-cools to well below the evaporation temperature of water. As a result, liquid water re-condenses and the soaked fuel grain will simply not ignite. This problem appears to be endemic to all peroxide hybrid rockets but is especially problematic when lower concentration peroxides (<90%) are used.

[0053] The ESA-funded Nammo Raufoss Project has been ongoing since 2003 and is currently the most accomplished of the existing peroxide-hybrid programs. The Nammo design used HTPB as the accompanying fuel. For this approach the peroxide solution was decomposed using SAAB's proprietary catalyst bed design, with the resulting hot gasses injected into the combustion chamber through a vortex injector. The catalyst bed hybrid was able to work with peroxide concentrations as low as 87.5%, but the catbed was quite large and made up a considerable fraction of the overall inert motor weight.

[0054] For the typical Nammo motor ignition sequence, after peroxide flow is initiated the chamber pressure gradual builds up from ambient to a plateau at approximately 1500 kPa (220 psia). This smoldering buildup of chamber pressure takes slightly more than 2 seconds, followed by a sharp rise in chamber pressure to approximately 2500 kPa bars (360 psia). NAMMO refers to the initial pressure buildup as the “mono-propellant combustion mode.” and the sharp rise and subsequent plateau as “hybrid combustion mode.”

[0055] As reported by Whitmore and Merkley (2017), no matter the concentration level, the initial expansion from the catbed exit to ambient will super-cool the water vapor in the decomposition products and result in a wet motor. This expansion and adiabatic cooling phenomenon is likely the reason for the large ignition latencies and the self-described monopropellant combustion modes, smoldering burns, and large ignition latencies experienced by the NAMMO hybrid motors. In any case, the resulting enthalpy levels are often too low to achieve full combustion, and ignition is highly unreliable.

[0056] In addition to the previously-described ignition reliability issues, catalyst beds also pose a series of operational issues: catbeds are heavy and volumetrically inefficient. They contribute nothing to the propulsive mass of the system. In order to be effective, catalysts are externally heated to high temperatures, often exceeding 300° C. As described previously, this pre-heat presents a serious problem for SmallSats that have limited energy budgets. Catbeds often self-consume at the high temperatures necessary for efficient decomposition action. Catalyst beds can be poisoned and rendered ineffective in the presence of stabilizers in HTP.

[0057] Because H.sub.2O.sub.2 catbeds tend to rely on noble metals like silver or platinum as the active agent, they are extremely expensive, and the final two events in the above list can significantly increase programmatic development costs.

[0058] Previous work by Whitmore and Martinez have demonstrated that a catalytic-assist, where a catalyst bed is placed in line with the system, significantly increases ignition reliability and reduces ignition latency. Catalytic-assist works by partially decomposing the incoming oxidizer to release free oxygen before entering the combustion chamber. Because catalytic-assist is only intended to assist the arc ignition system, and not fully decompose the incoming HTP; pre-heat is not required, and far less expensive materials like potassium permanganate, manganese dioxide, manganese (III) oxide, and potassium nitrate can be used in lieu of silver and platinum.

[0059] Even with the success of the previously described catalytic-assist methods, considering the issues associated with catbeds, including weight and volume, it is desirable to remove the catalytic system from the design. An alternative method initiates combustion using a gaseous oxygen pre-

lead and then introduces HTP to the hot combustion chamber. Residual energy from the GOX/ABS combustion thermally decomposes the HTP flow, with the freed oxygen allowing full hybrid combustion to initiate.

[0060] Previously, this thermal ignition method was applied by Whitmore for ignition of a larger-76 mm, 140 N thrust hybrid system. Thermal decomposition for an HTP-hybrid on the proposed 1 N to 5 N thrust level used for SmallSat propulsion was investigated by Smith. While he saw success with this method, there are still drawbacks for this system. For example, system architecture doubles in complexity because separate plumbing, valves, and sensors are used for the GOX lines. Further, spacecraft volume is sacrificed because transitioning from a ground-based developmental unit to a space-ready flight unit requires a high pressure GOX tank and its associated plumbing and controls, which do not directly contribute to the propulsion system's power or efficiency. The smaller the spacecraft is, the greater the percent volume increase, making the drawback more significant. For a spacecraft classified as a Small Satellite, this approach is not viable to allow for enough HTP propellant to be stored onboard to achieve a delta-V useful for most space missions.

[0061] Tests indicate that potassium permanganate (KMnO₄) can be heterogeneously infused into a 3D printed ABS shell, significantly enhancing ignition-reliability without adversely affecting the overall system performance. Here the infused catalyst initiates HTP oxidizer decomposition upon contact, releasing oxygen gas and enabling the effectiveness of the arc-ignition system. Because infused catalyst directly releases oxygen in the combustion chamber, the need for an external catbed or thermal pre-lead is eliminated.

[0062] FIG. 3 is a perspective view of an example infused solid polymeric fuel 300 as described herein. This example is a fuel grain for use in a hybrid rocket. In other examples, the infused solid polymeric fuel can be an explosive. This example fuel grain is cylindrically shaped with a cylindrical bore 310 in the center. The fuel grain can be in the form of a polymeric body. In some examples, the polymeric body can be made from most common 3D printing plastics. In some examples, the polymer can include, but is not limited to, acrylonitrile butadiene styrene (ABS), poly(methyl methacrylate) (PMMA), polyamide, Nylon, polyphenyl ether (PPE), polyethylene (PE), low-density polyethylene (LDPE), high-density polyethylene (HDPE), or combinations thereof. A spark cap 312 is also shown which can be associated with the solid polymeric fuel 300 to provide ignition of the fuel during use and is described in more detail with respect to FIG. 5.

[0063] FIG. 4A shows a magnified cross-section of a polymeric body 400. The polymeric body can be built up from fused layers of filament 410, with void spaces 420 being left between strands of filament. In some cases, these void spaces 420, or pores, can be continuously distributed throughout a volume of the polymeric body. The void spaces can also be connected to the external surface of the polymeric body. This porous polymeric body can have a porosity from about 1% to about 50%, meaning that from about 1% to 50% of the geometric volume of the porous polymeric body is void space. Porosity values can depend on the propellant combinations selected but also depend on several other factors such as chamber pressure, desired regression rate/thrust, or plume characteristics, for example. In further examples, the porosity can be from 1% to 30%, or from 1% to 20%, or from 1% to 10%, or from 1% to 5%, or from 5% to 50%, or from 5% to 30%, or from 5% to 20%, or from 5% to 10%.

[0064] In some examples, a combustion enhancement agent can be infused onto the polymeric body. The combustion enhancement agent can be deposited on interior surfaces of the pores and on the external surface of the polymeric body. In some examples, the combustion enhancement agent can be a catalyst for decomposing an oxidizer such as hydrogen peroxide. Because the catalyst is present on the surfaces of the polymer (as opposed to embedded within the polymer), the catalyst is readily available to decompose hydrogen peroxide that comes in contact with the infused polymeric fuel. This allows the fuel and oxidizer to be ignited more quickly and easily. For this reason, the infusion process described herein can allow for better ignition characteristics compared to other

processes that involve melting the polymer and mixing the catalyst into the molten polymer. In other such processes, the catalyst is not as readily available at the surface and the reaction of the catalyst with the oxidizer does not occur as quickly. In further examples, the combustion enhancement agent can be a catalyst, an oxidizer, a hypergolic fuel, or a combination thereof. [0065] In some examples, the combustion enhancement agent can be water-soluble. In other examples, the combustion enhancement agent can be insoluble to water. Additionally, the combustion enhancement agent can be soluble in a solvent that does not dissolve the porous polymeric body. In examples that utilize an insoluble combustion enhancement agent, the combustion enhancement agent can be in the form of nanoparticles or microparticles that are small enough to enter the porous in the porous polymeric body. In some examples, the combustion enhancement agent can include particles with a particle size less than 100 μm , or less than 10 μm , or less than 1 μm . Some example combustion enhancement agents that may be present as solid particles can include metals such as platinum, silver, ruthenium, rhodium, iridium, palladium, copper, iron, nickel, or others.

[0066] In various examples, the combustion enhancement agent can be a catalyst such as potassium permanganate, sodium permanganate, potassium iodide, iron chloride (FeCl_3), copper sulfate, manganese oxide, platinum, silver, iron oxide (Fe_2O_3), lead oxide (PbO), ruthenium, rhodium, iridium, palladium, copper, iron, nickel, or combinations thereof. In other examples, the combustion enhancement agent can be an oxidizer such as potassium perchlorate, potassium nitrate, sodium perchlorate, sodium hypochlorite, hydroxyl ammonium nitrate, ammonium perchlorate, ammonium nitrate, ammonium dinitramide, ammonium chlorate, lithium nitrate, or combinations thereof. In further examples, the combustion enhancement agent can be a hypergolic fuel such as sodium borohydride.

[0067] Further, using this method solid-propellants can be printed and infused with varying degrees of oxidizer and fuel components. In this process, depending upon the blended propellant's oxidizer-of-fuel ratio (O/F), the resulting infusion can act as a stand-alone-propellant solid propellant, or a hybrid fuel source offsetting the required external oxidizer, allowing lower mass flow for the associated fluidic oxidizing agent. Using this process, an energetic material can be made reversibly switchable between a safe state (highly insensitive but low energetic yield) and a performance state (high energetic yield but sensitive).

[0068] The present disclosure also describes methods of making infused solid polymeric fuels. In some examples, the infused solid polymeric fuel can be a fuel grain for a hybrid rocket. In other examples, the infused solid polymeric fuel can be an explosive. The method can include forming a porous polymeric body. The porous polymeric body can be formed by additive manufacturing. The additive manufacturing can include fused deposition modeling (FDM) 3D printing, which is a material extrusion method of additive manufacturing where materials are extruded through a nozzle and joined together to create 3D objects. In particular, the standard FDM process distinguishes itself from other material extrusion techniques by using thermoplastics as feedstock materials. In the present disclosure, the polymeric body is built up from fused layers of filament. During this process, the FDM 3D printing can be performed with different infill densities. In some examples, the infill density can be from about 25% to about 50%, or from about 30% to about 75%, or from about 40% to about 80%, or from about 50% to about 100%. In some examples, no matter what the infill density is, there can be void spaces in between filaments, with these void spaces, or pores, being continuously connected. These continuous pores can be distributed throughout the volume of the polymeric body and can be connected to the external surface of the polymeric body. The porous polymeric body can have a porosity from about 1% to about 50%. In further examples, the porosity can be from 1% to 30%, or from 1% to 20%, or from 1% to 10%, or from 1% to 5%, or from 5% to 50%, or from 5% to 30%, or from 5% to 20%, or from 5% to 10%. In some examples, the polymer used to make the polymeric body can include, but is not limited to, acrylonitrile butadiene styrene (ABS), poly(methyl methacrylate) (PMMA), polyamide, Nylon, polyphenyl ether (PPE),

polyethylene (PE), low-density polyethylene (LDPE), high-density polyethylene (HDPE), or combinations thereof.

[0069] The methods described can also include soaking the porous polymeric body in a composition. The composition can include a combustion enhancement agent. In some examples, the composition can also include a solvent to help the combustion enhancement agent reach the pores inside the porous polymeric body. The composition can comprise the combustion enhancement agent dissolved in the solvent. The concentration of the combustion enhancement agent in the composition can be from about 1 wt % to about 15 wt %, from about 2 wt % to about 20 wt %, or from about 1 wt % to about 12 wt %. Soaking the porous polymeric body results in a deposit of the combustion enhancement agent on interior surfaces of the pores and on the external surface of the polymeric body. As previously mentioned, the void spaces can be continuously connected and accessible through pores in the surface. Therefore, when the porous polymeric body is soaked in a solution of a combustion enhancement agent, the solution can reach the void spaces inside the porous polymeric body.

[0070] In some examples, the combustion enhancement agent can be water soluble. The composition used for soaking the porous polymeric body can include water and the combustion enhancement agent dissolved in the water. Some polymers that can be used to make the porous polymeric body can be hydrophobic. Therefore, it may be difficult for water to flow into the small pores of the polymeric body because of the hydrophobic nature of the polymer. Increased pressure can be used to push the aqueous solution into the pores of the porous polymeric body. Therefore, in some examples, the soaking can be performed under a pressure above atmospheric pressure. The pressure can be from about 50 psi to about 150 psi, or from about 75 psi to about 150 psi, or from about 100 psi to about 150 psi, or from about 50 psi to about 100 psi, or from about 75 psi to about 100 psi. The porous polymeric body can be placed into a vessel filled with the solution of the combustion enhancing agent and the vessel can be pressurized. The porous polymeric body can then be soaked in the solution under pressure for a sufficient time to allow the combustion enhancement agent to diffuse into the porous polymeric body. In some examples, the soaking time can be from about 1 hour to about 5 days, or from about 1 hour to about 2 days, or from about 1 hour to about 1 day, or from about 1 hour to about 12 hours, or from about 12 hours to about 5 days, or from about 12 hours to about 2 days.

[0071] Soaking under pressure for these soaking times can also be useful when the composition containing the combustion enhancement agent includes a solvent other than water, and when the combustion enhancement agent includes insoluble solid particles. In some examples, the soaking process can include mixing the composition of the solvent and the combustion enhancement agent during soaking. In a certain example, the mixing can be gentle, such as by rotating the vessel containing the composition and the porous polymeric body. In some examples, the vessel can be rotated at a speed from 1 rpm to 30 rpm. The rotation axis can be vertical or horizontal in some examples. In examples that utilize an insoluble combustion enhancement agent, the combustion enhancement agent can be in the form of nanoparticles or microparticles that are small enough to enter the porous in the porous polymeric body. In some examples, the combustion enhancement agent can include particles with an average particle size less than 100 μm , or less than 10 μm , or less than 1 μm . Some example combustion enhancement agents that may be present as solid particles can include metals such as platinum, silver, ruthenium, rhodium, iridium, palladium, copper, iron, nickel, or others.

[0072] In various examples, the combustion enhancement agent can be a catalyst such as potassium permanganate, sodium permanganate, potassium iodide, iron chloride (FeCl_3), copper sulfate, manganese oxide, platinum, silver, iron oxide (Fe_2O_3), lead oxide (PbO), ruthenium, rhodium, iridium, palladium, copper, iron, nickel, or combinations thereof. In other examples, the combustion enhancement agent can be an oxidizer such as potassium perchlorate, potassium nitrate, sodium perchlorate, sodium hypochlorite, hydroxyl ammonium nitrate,

ammonium perchlorate, ammonium nitrate, ammonium dinitramide, ammonium chlorate, lithium nitrate, or combinations thereof. In further examples, the combustion enhancement agent can be a hypergolic fuel such as sodium borohydride.

[0073] Furthermore, the methods described can include evaporating the solvent. FIG. 4B shows the cross-section of the porous polymeric body after it has been soaked in a solution of combustion enhancement agent and then dried to remove the solvent. This leaves a layer of combustion enhancement agent **430** on the exterior surface of the porous polymeric body and on the interior surfaces in the void spaces within the porous polymeric body. Evaporating can be done at room temperature or an elevated temperature. In some examples, the drying temperature can be from 60° C. to 80° C. The drying time can be from 1 hour to 24 hours, or from 2 hours to 12 hours, or from 4 hours to 12 hours, or from 6 hours to 12 hours, or from 6 hours to 8 hours, in some examples.

[0074] The present disclosure further describes a hybrid rocket system. The hybrid rocket system can include an infused solid polymeric fuel grain. The infused solid polymeric fuel grain can include a porous polymeric body having continuous pores distributed throughout a volume of the polymeric body and connecting to an external surface of the polymeric body. In some examples, the polymer can include, but is not limited to, acrylonitrile butadiene styrene (ABS), poly(methyl methacrylate) (PMMA), polyamide, Nylon, polyphenyl ether (PPE), polyethylene (PE), low-density polyethylene (LDPE), high-density polyethylene (HDPE), or combinations thereof.

[0075] The infused solid polymeric fuel can also have a decomposition catalyst infused into the polymeric body. Non-limiting examples of suitable decomposition catalyst can include potassium and sodium-based salts, noble metals, and copper-oxides. The catalyst can be deposited on interior surfaces of the pores and on the external surface of the polymeric body. The hybrid rocket system can also include a liquid oxidizer injector operable to contact the fuel grain with liquid oxidizer such that the catalyst catalyzes a decomposition of the liquid oxidizer. In some examples, the liquid oxidizer can be hydrogen peroxide, nitrous oxide, various less-stable mixtures-of-nitrides (MON), or a combination thereof. The system can also include an igniter operable to ignite the infused solid polymer fuel grain. In this process the ignition path can be directly 3-D printed into the propellant materials.

[0076] In some examples, the oxidizer can have a concentration less than 90%, or from 90% to less than 95%, or 95% or greater. In further examples, the igniter comprises an electric arc ignition system. The igniter can apply less than 5 W of power to ignite the infused solid polymer fuel grain in some examples. In further examples, the igniter can apply less than 5 Joules of energy to ignite the infused solid polymer fuel grain.

Examples

[0077] FIG. 5 shows a specific motor configuration, which includes the injector cap **510** containing a COTS spray injector nozzle **520**, 38 mm motor casing **530**, ignition cap, fuel grain **540**, and 1 N nozzle retainer assembly **550**. The motor case **530** was machined from 316-grade stainless steel. The injector cap **510**, which includes ports for the injector **560**, ignition electrodes **570** and chamber pressure fitting, was machined from 6061-T6 grade aluminum. The nozzle **520** was machined from a single piece of graphite. The spark cap **580** was printed separately from the fuel grain **540** to ensure that the catalyst infusion does not inhibit the arcing capabilities of the ABS to achieve a strong spark for ignition.

[0078] FIG. 6 shows the test cart used for the tests conducted and the system Piping and Instrumentation diagram associated with the layout of the cart. The system is pressurized with Nitrogen gas (N₂) or GOX to dictate HTP flow rates. A separate path for N₂ leads directly to motor injection to provide a fire suppression and purge function in the case of a misfire. Fire control and data acquisition are managed from outside the test cell using laptop computer that communicates with the instrumentation system via a single Ethernet Bus. The GOX and HTP valves and ignition system are powered separately through arming switches, giving each system a “safe” and “armed” state to eliminate the possibility of accidental ignition. Motor performance measurements include

motor injector and chamber pressure, and thrust level is sensed by a load cell mounted to the test sled.

[0079] As shown in FIG. 7, a test mounting assembly which includes a thruster chamber **910**, a thrust stand **920**, multiple flexures **930**, and a load cell **940**. The thruster chamber **910** is mounted on an inverted pendulum thrust stand **920**. This stand includes an aluminum T-slot railing supported by three thin aluminum flexures **930**. These flexures allow the structure to move in the direction of the thrust axis when the motor fires, resulting in load cell **940** measurements. In the illustrated configuration, the motor fires to the right of the test sled in this image.

[0080] The CatGrain is created by diffusing a potassium permanganate (KMnO₄) catalyst solution into a fuel grain. To achieve sufficient diffusion, the fuel grain material can be porous. Most fuel grains used for hybrid rocket motors are plastic, which are not porous enough as extruded or cast material. Plastic filament used for FDM 3D printing is porous, and the layering in printed parts provides additional locations for diffusion. Printed ABS appears to be a good choice for this application because of the porosity and burn characteristics with HTP.

[0081] The chemical reaction between KMnO₄ and HTP is



[0082] Although KMnO₄ is not a true catalyst for HTP (KMnO₄ reduces to Manganese Oxide (MnO₂) as a purer catalyst form as shown in the chemical balance equation above), MnO₂ is not soluble in a solution that does not break down ABS plastic, whereas KMnO₄ is soluble. For convenience and simplicity, water is used as the solvent for the catalyst solution.

[0083] Preliminary results from the initial testing campaign include a total of 14 successful full-combustion CatGrain hot firings. From the initial validation tests conducted, the CatGrain shows strong promise, with initial test data shown in FIGS. **8A-8D**. A rise time of 1.3 seconds is observed, with average steady-state ambient Isp and c* performance values at 177 seconds and 1234 m/s, respectively. The O/F ratio is lower than ideal. This is due to the KMnO₄ increasing combustion pressure more than a similar setup with a non-catalyzed ABS fuel grain typically achieves, which reduces the pressure-fed HTP injection, resulting in a fuel rich combustion environment. Additionally, the increase in chamber pressure also increased the generated thrust.

[0084] Core burning hybrid motors are susceptible to an O/F ratio shift during the burn, especially the first several seconds of the initial burn when the combustion port area changes most rapidly. This is evident in this test, where the O/F ratio starts at around 2 but decays down to 1 over the 15 second hot-fire. The result is a less stoichiometric burn, yielding lower thrust and therefore reduced performance than the theoretical peak. Despite this, the performance attained from this initial test shows the potential for this system to match and outperform hydrazine and newer green derivatives.

[0085] For comparison, a test with a normal ABS fuel grain—not a CatGrain—was conducted. The results are shown in FIG. **9**. A significant rise time is seen, about 5.02 seconds. Because there was so short of time of actual combustion, there was very little mass burned. As such, the calculations to calculate the fuel mass flow and subsequent performance data could not be sufficiently anchored by the measured fuel burned. Thus, performance data is not presented, only the measured thrust.

[0086] An issue that results from the low O/F ratio is the difficulty to reignite the CatGrain without needing to re-dope the surface with more KMnO₄. The low O/F ratio produces a dirtier burn, leaving a significant amount of soot deposited on the burned surface. This inhibits the ability for the infused catalyst to directly contact HTP upon injection to release GOX to aid in ignition. Additionally, the erosive burning prevalent with an atomizing spray injector reported by Smith eventually burns away portions of the ignition cap, which eliminates the electrical path after extended burns.

[0087] Several configurations can prevent the spray injection from burning away the spark cap while also maintaining a source of unburned and soot-free catalyst to provide the GOX release

necessary for reliable ignition.

[0088] One method is to redesign the spark cap in a way that insulates the ABS ignition fuel with a high-temperature ceramic, shown in FIG. 3. Boron-Nitride was selected due to its high thermal resistance and being a porous ceramic. The porosity allowed for the BN insulator to absorb KMnO_4 and provide an additional source of HTP decomposition for GOX release for ignition.

[0089] Though this did work for the first hot fires, this method did not work long-term. The small geometry associated with this size of motor resulted in the ceramic insulator fracturing from the pressure and thermal shock, sending fragments down the chamber and clogging the nozzle.

[0090] The boron-nitride insulator was too brittle, a ceramic tape cloth was employed. A high-temperature alumina tape with a temperature rating to 1700°C . was also infused with KMnO_4 , formed into a small ring, and secured to the walls of the inner opening of the spark cap. This method worked better, but unfortunately was still partially consumed where the cloth meets the fuel grain combustion region, indicating the point where combustion temperatures reach above 1700°C . This provided successful insulation for the electrodes of the spark cap but did not allow for maintaining unconsumed KMnO_4 where it would directly impinge with HTP to release GOX for ignition.

[0091] Keeping the alumina tape insulation to protect the electrodes was still useful, but an additional component was used to provide the catalyst support that will not be consumed or coated by soot. A stainless-steel wire mesh disc was chosen, first tested with a gridded mesh of 20×20 holes per one square inch. The mesh was impregnated with KMnO_4 , then calcined in a furnace at 600°C . for 2 hours to burn away the potassium and yield a manganese oxide (MnO_2) deposit, which is a true catalyst for HTP. This means that there is no redox reaction like there is with KMnO_4 and thus should not be reduced with the volume of HTP being passed through the mesh.

[0092] The 20×20 mesh grid was chosen to allow for some HTP to contact with impregnated mesh while some to pass through undecomposed. The mesh is placed on top of the spark cap, coming in direct contact with the injector exit. This configuration yields a “staged decomposition” by having multiple points for initial decomposition of fractions of the HTP volume before sustained combustion is reached.

[0093] This method proved quite effective, as the physical separation from the combustion region within the CatGrain helped reduce the buildup of soot, though not completely eliminate it. Upon inspection post-fire, there was still small amounts of soot coating the mesh. However, when the soot coating was brushed off and a drip test was performed on the mesh, it demonstrated a retention in strong catalytic activity, proving the usefulness of calcining the KMnO_4 to yield MnO_2 .

[0094] Though the stainless-steel mesh method showed initial success, the nature of this material does not allow for much infusion of catalyst into the steel. A thin metal foam sheet would provide a better structure to support more catalyst material, potentially further improving the ignition latency. Nickel foam was chosen because it has a higher melting temperature and is cheaper to produce.

[0095] The pore density of metal foam absorbed more catalyst, yielding about 10% MnO_2 as compared to 2.5% for the stainless-steel mesh. The increased pore density proved to not be useful, as it slowed down the oxidizer flow slightly and also served to push some HTP more radially, increasing the erosive burning on the spark cap.

[0096] To avoid the reduction in oxidizer flow through the nickel foam, several small holes were drilled through the center to act more like the wire mesh, where some peroxide could pass through unobstructed and undecomposed. This modification drastically improved the ignition latency down to 0.741 seconds, shown in FIG. 10, though did not fully get rid of the radial flow issue.

Additionally, the nickel foam material was only 0.3 mm thick, which ended up getting burned through preventing use for re-ignition. This is likely from erosion burning from the HTP instead of the actual combustion temperature in the chamber. A thicker foam would survive better but may not

allow for staged decomposition. This has yet to be tested.

[0097] From these findings, there appears to be an optimal spacing or pore size for the staged decomposition ignition method while retaining structure and catalyst to attain re-ignition.

[0098] The last 10 tests all successfully achieved ignition, with ignition latencies and associated ignition methods reported in Table 1. To date, re-ignition has not reliably been proven, largely due to having burns that are too fuel rich that leave soot deposits within the chamber. CatGrains that were burned more than once were ignited with a new alumina tape or metal mesh.

TABLE-US-00001 TABLE 1 Rise Time Date (seconds) Configuration Mar. 23, 2023 1.6 CatGrain4 with Boron Nitride Apr. 6, 2023 1.859 CatGrain5 with Alumina Tape Apr. 17, 2023 3.549 CatGrain6 with Alumina Tape Apr. 26, 2023 1.503 CatGrain6 with Alumina Tape May 3, 2023 1.481 CatGrain6 with Alumina Tape May 19, 2023 1.242 CatGrain8, Alumina Tape, SS Mesh May 19, 2023 0.926 CatGrain8, Alumina Tape, SS Mesh May 30, 2023 1.87 CatGrain10, Nickel Foam May 30, 2023 0.741 CatGrain10, Alumina Tape, Nickel Foam May 30, 2023 0.616 CatGrain10, Alumina Tape, Nickel Foam

[0099] The CatGrain system was used as a true drop-in replacement, allowing HTP to be burned directly in the legacy 1-N GOX/ABS HPGHP thruster with no hardware modifications. However, because of the longer aspect ratio associated with the legacy GOX/ABS thrust chamber, the resulting O/F ratios for HTP/CatGrain were well-below the optimal point for HTP combustion; thus, the associated specific impulse values are lower than would be achieved for an optimized geometry, varying between 170 and 185 seconds. Assuming a high expansion ratio nozzle (25:1), these values extrapolate to between 225-250 seconds for vacuum specific impulse,

[0100] Even with the non-optimal thrust chamber geometry, the CatGrain/HPGHP configuration still out-performs monopropellant hydrazine and the ionic-liquid green hydrazine alternatives. There are clear steps that can be taken to improve the O/F ratio of this CatGrain hybrid motor, starting with shortening the fuel grain length.

[0101] The foregoing detailed description describes the invention with reference to specific exemplary embodiments. However, it will be appreciated that various modifications and changes can be made without departing from the scope of the present invention as set forth in the appended claims. The detailed description and accompanying drawings are to be regarded as merely illustrative, rather than as restrictive, and all such modifications or changes, if any, are intended to fall within the scope of the present invention as described and set forth herein.

Claims

1. An infused solid polymeric fuel, comprising: a porous polymeric body having continuous pores distributed throughout a volume of the polymeric body and connecting to an external surface of the polymeric body; and a combustion enhancement agent infused into the polymeric body, wherein the combustion enhancement agent is deposited on interior surfaces of the pores and on the external surface of the polymeric body, wherein the combustion enhancement agent comprises a catalyst, an oxidizer, a hypergolic fuel, or a combination thereof.
2. The infused solid polymeric fuel of claim 1, wherein the infused solid polymeric fuel is a fuel grain for a hybrid rocket.
3. The infused solid polymeric fuel of claim 1, wherein the infused solid polymeric fuel is an explosive.
4. The infused solid polymeric fuel of claim 1, wherein the polymeric body comprises a polymer selected from the group consisting of acrylonitrile butadiene styrene (ABS), poly(methyl methacrylate) (PMMA), polyamide, Nylon, polyphenyl ether (PPE), polyethylene (PE), low-density polyethylene (LDPE), high-density polyethylene (HDPE), and combinations thereof.
5. The infused solid polymeric fuel of claim 1, wherein the polymeric body has a porosity from about 1% to about 50%.

6. The infused solid polymeric fuel of claim 1, wherein the combustion enhancement agent is water-soluble.
7. The infused solid polymeric fuel of claim 1, wherein the combustion enhancement agent is soluble in a solvent that does not dissolve the porous polymeric body.
8. The infused solid polymeric fuel of claim 1, wherein the combustion enhancement agent comprises insoluble solid nanoparticles or microparticles.
9. The infused solid polymeric fuel of claim 1, wherein the combustion enhancement agent is a catalyst selected from the group consisting of potassium permanganate, sodium permanganate, potassium iodide, iron chloride (FeCl₃), copper sulfate, manganese oxide, platinum, silver, iron oxide (Fe₂O₃), lead oxide (PbO₂), ruthenium, rhodium, iridium, palladium, copper, iron, nickel, and combinations thereof.
10. The infused solid polymeric fuel of claim 1, wherein the combustion enhancement agent is an oxidizer selected from the group consisting of potassium perchlorate, potassium nitrate, sodium perchlorate, sodium hypochlorite, hydroxyl ammonium nitrate, ammonium perchlorate, ammonium nitrate, ammonium dinitramide, ammonium chlorate, lithium nitrate, or combinations thereof.
11. The infused solid polymeric fuel of claim 1, wherein the combustion enhancement agent is sodium borohydride.
12. A method of making an infused solid polymeric fuel, comprising: forming a porous polymeric body having continuous pores distributed throughout a volume of the polymeric body and connecting to an external surface of the polymeric body; soaking the porous polymeric body in a composition comprising a combustion enhancement agent and a solvent to deposit the combustion enhancement agent on interior surfaces of the pores and on the external surface of the polymeric body; and evaporating the solvent, wherein the combustion enhancement agent comprises a catalyst, an oxidizer, a hypergolic fuel, or a combination thereof.
13. The method of claim 12, wherein forming the porous polymeric body is performed by additive manufacturing.
14. The method of claim 13, wherein the additive manufacturing comprises fused deposition modelling (FDM) 3D printing with an infill density from 50% to 100%.
15. The method of claim 12, wherein the soaking is performed under a pressure above atmospheric pressure.
16. The method of claim 15, wherein the pressure is from 50 psi to 150 psi.
17. The method of claim 12, wherein a concentration of the combustion enhancement agent in the composition is from 1 wt % to 15 wt %.
18. The method of claim 12, wherein the composition comprises the combustion enhancement agent dissolved in the solvent.
19. The method of claim 12, wherein the composition comprises solid particles of the combustion enhancement agent mixed with the solvent, and wherein the solid particles have an average particle size less than 100 μm.
20. The method of claim 12, wherein the infused solid polymeric fuel is a fuel grain for a hybrid rocket or is an explosive.
21. The method of claim 12, wherein the polymeric body comprises a polymer selected from the group consisting of acrylonitrile butadiene styrene (ABS), poly(methyl methacrylate) (PMMA), polyamide, Nylon, polyphenyl ether (PPE), polyethylene (PE), low-density polyethylene (LDPE), high-density polyethylene (HDPE), and combinations thereof.
22. The method of claim 12, wherein the polymeric body has a porosity from about 1% to about 50%.
23. The method of claim 12, wherein the combustion enhancement agent is at least one of: a catalyst selected from the group consisting of potassium permanganate, sodium permanganate, potassium iodide, iron chloride (FeCl₃), copper sulfate, manganese oxide, platinum, silver, iron oxide (Fe₂O₃), lead oxide (PbO₂), ruthenium, rhodium, iridium, palladium,

copper, iron, nickel, and combinations thereof; an oxidizer selected from the group consisting of potassium perchlorate, potassium nitrate, sodium perchlorate, sodium hypochlorite, hydroxyl ammonium nitrate, ammonium perchlorate, ammonium nitrate, ammonium dinitramide, ammonium chlorate, lithium nitrate, or combinations thereof; and sodium borohydride.

24. A hybrid rocket system, comprising: an infused solid polymeric fuel grain, comprising: a porous polymeric body having continuous pores distributed throughout a volume of the polymeric body and connecting to an external surface of the polymeric body, and a decomposition catalyst infused into the polymeric body, wherein the catalyst is deposited on interior surfaces of the pores and on the external surface of the polymeric body; a liquid oxidizer injector operable to contact the fuel grain with liquid oxidizer such that the catalyst catalyzes a decomposition of the liquid oxidizer; and an igniter operable to ignite the infused solid polymeric fuel grain.

25. The system of claim 24, wherein the liquid oxidizer has a concentration less than 90%.

26. The system of claim 24, wherein the igniter comprises an electric arc ignition system, wherein the igniter applies less than 5 W of power to ignite the infused solid polymer fuel grain, or wherein the igniter applies less than 5 Joules of energy to ignite the infused solid polymer fuel grain.

27. The system of claim 24, wherein the liquid oxidizer is hydrogen peroxide, nitrous oxide, or a combination thereof.
